A positive premagic matrix that is not diagonalizable

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Abstract

Conjecture 7.2 of Teknomo's *Premagic and Ideal Flow Matrices* asserts that every premagic matrix is diagonalizable. We give a positive integer counterexample of order three, the least possible real order. Its zero eigenvalue has algebraic multiplicity two and geometric multiplicity one. Scaling gives a positive doubly stochastic counterexample; adding a positive multiple of the identity gives invertible ones.

Task 1: the original conjecture

Let $\mathbf{1}$ be the column vector of ones. A real square matrix A is *premagic* when its vector of row sums equals its vector of column sums, that is,

$$A\mathbf{1} = A^T\mathbf{1}.$$

This is Definition 4.3 of the final article [1], page 37. Conjecture 7.2, page 44, asserts that such a matrix is diagonalizable and has a full set of independent eigenvectors. The same conjecture appears in the current arXiv manuscript. The definition imposes no zero-diagonal condition. Our example is strictly positive, so it also satisfies nonnegativity and irreducibility if these properties are additionally desired in the ideal-flow setting.

Theorem 1. *The positive integer matrix*

$$M = \begin{pmatrix} 4 & 4 & 1 \\ 2 & 2 & 5 \\ 3 & 3 & 3 \end{pmatrix}$$

is premagic but is not diagonalizable, even over $\mathbb{C}$.

Proof. Every row sum and every column sum is 9. Thus M is premagic. Put

$$j = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, \qquad u = \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}, \qquad v = \begin{pmatrix} 1 \\ 1 \\ -2 \end{pmatrix}.$$

These vectors form a basis: the matrix $B = (j\ u\ v)$ has determinant 6. Direct multiplication gives

$$Mj = 9j, \qquad Mu = 0, \qquad Mv = 6u.$$

Consequently

$$B^{-1}MB = \begin{pmatrix} 9 & 0 & 0 \\ 0 & 0 & 6 \\ 0 & 0 & 0 \end{pmatrix}.$$

The characteristic polynomial is $x^2(x-9)$. The eigenspace for 0 is exactly $\mathbb{C}u$ and that for 9 is exactly $\mathbb{C}j$. There are only two independent eigenvectors, although the dimension is three. This disproves both assertions of the conjecture. □

The construction can also be written $M = 3jj^T + uv^T$. The equalities $j^T u = j^T v = v^T u = 0$ make uv^T a nonzero square-zero operator on $j^\perp$, while $3jj^T$ acts by 9 on j and vanishes on $j^\perp$. Hence the positive dominant eigenvalue is simple, exactly as expected from Perron–Frobenius theory; the defect occurs at the other eigenvalue.

Proposition 1. *The matrix $M/9$ is a strictly positive doubly stochastic matrix that is not diagonalizable. For every real $t > 0$, the matrix $M + tI$ is strictly positive, invertible, premagic, and not diagonalizable. No real premagic matrix of order less than three is defective.*

Proof. Dividing by 9 makes all row and column sums one and preserves the eigenspaces and the nontrivial Jordan block. Adding tI changes the eigenvalues to $9 + t, t, t$ and preserves the same block and eigenvectors; its determinant is $(9 + t)t^2 > 0$ and its row and column sums are $9 + t$. For order two, write $A = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$. The premagic condition gives $a + b = a + c$, so $b = c$. Thus A is real symmetric and diagonalizable. Order one is immediate. □

Verification and scope

The proof uses only exact matrix identities. The accompanying standard-library Python verifier checks the row and column sums, $\det B = 6$, the similarity relation $MB = BJ$, the characteristic polynomial coefficients, and the scaled and shifted identities using integer and rational arithmetic. No floating-point eigensolver is used. This resolves MathDB 336488 as a counterexample to the full universal statement. The minimal-order assertion here concerns real matrices. The source and bounded prior-claim search are documented separately. The review is by the author agent, without independent peer review.

Four follow-up tasks

These are proposed problems, not additional solved credits.

Task 2. Determine the least total entry sum of a strictly positive integer 3×3 premagic matrix that is not diagonalizable over $\mathbb{C}$, and classify all minimizers up to simultaneous permutation of rows and columns.

Task 3. Determine the least order of a nonnegative, irreducible, zero-diagonal premagic matrix that is not diagonalizable over $\mathbb{C}$, and provide an exact counterexample at that order or prove that no such matrix exists.

Task 4. For every $n \geq 3$, characterize the lists of complex eigenvalues and Jordan block sizes realizable by strictly positive doubly stochastic $n \times n$ matrices.

Task 5. Give an explicit parametrization of all defective real 3×3 premagic matrices, including necessary and sufficient inequalities on the parameters for every entry to be strictly positive.

References

[1] K. Teknomo, *Premagic and Ideal Flow Matrices*, Data Science: Journal of Computing and Applied Informatics **3**(1) (2019), 35–45. doi:10.32734/jocai.v3.i1-621. Preprint: arXiv:1706.08856v2.